

Lumina

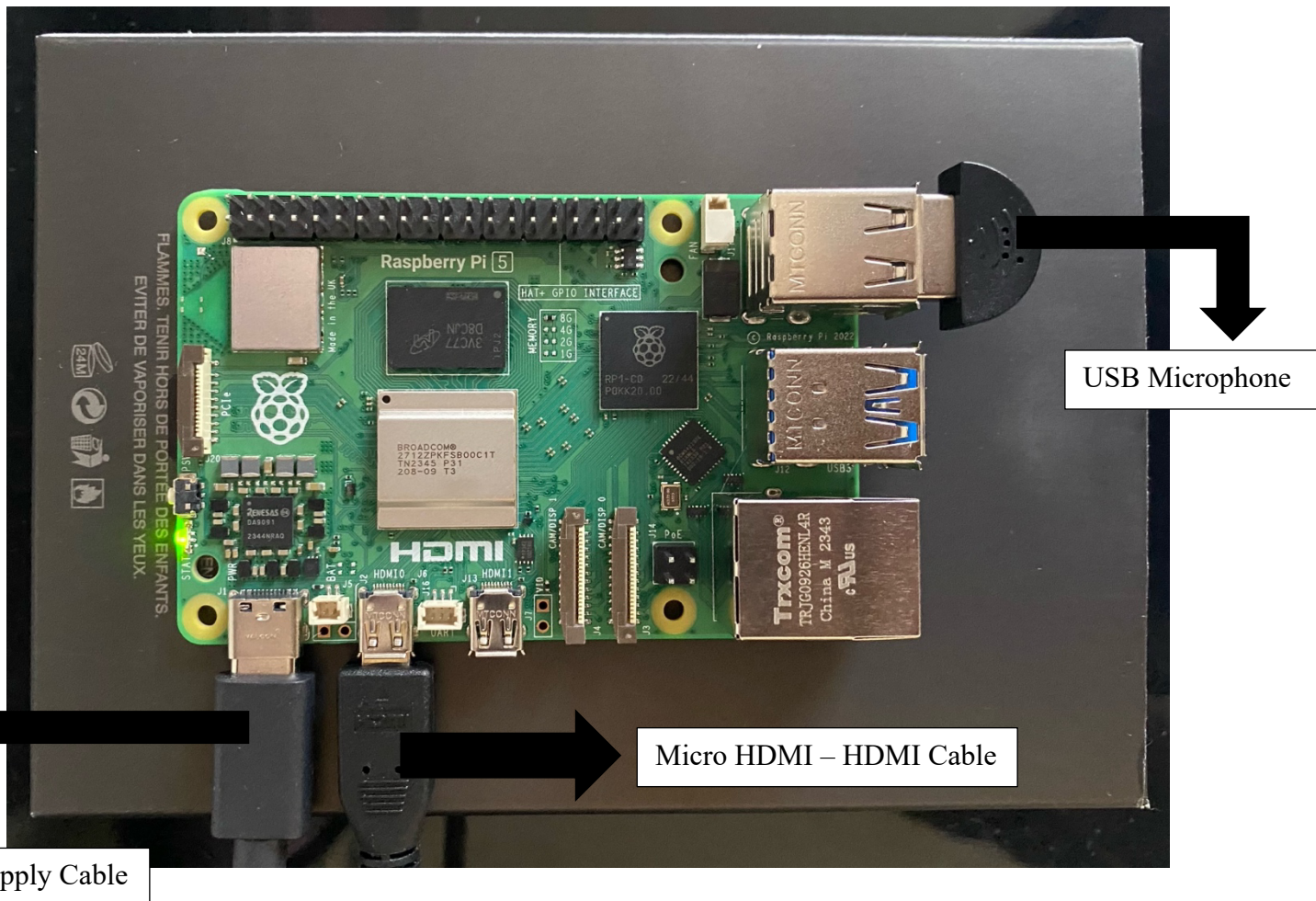
AI Art Generator

Our project uses voice control to create cool artwork and displays it on any TV through its HDMI output.

How does it work?

We used a python program that uses the Picovoice package to detect and understand user prompts spoken to Lumina and then send the prompts to DALL-E 3 through OpenAI's python library to be converted to an image that is then displayed on the screen. You can find the entire source code at the very end of the file.

A look at Lumina



Raspberry Pi 5: Raspberry Pi 5 provides the backbone for Lumina. It executes the software application that processes user input responses.

USB Microphone: The USB microphone is connected to one of the USB ports of Raspberry Pi 5. It receives the audio input from the user and conveys it to Raspberry Pi 5 for execution.

Micro HDMI to HDMI Cable: The micro HDMI port on the Raspberry Pi 5 is connected to the HDMI port on the display screen through this cable. This also allows the program we designed to identify the screen ratio and adapt itself to automatically.

See Lumina in Action

Lumina can display any request asked by the user. Let's see some here:

Prompt 1 – Lady in city walking by a park



Prompt 2 – Boy kicking around a blue football



Prompt 3 – professor teaching computer architecture in university



[Watch a video of Lumina in Action](#)

Phases of Project Completion

It was not easy to do this project, our main issues were not even with the actual coding, which we, as Computer Science students, found pretty easy. We actually struggled most with actually connecting the Raspberry Pi to our computers to program on. We solved this issue by making some changes to 'bashrc' file of the raspberry pi which basically allows us to customize the shell environment due to issues with the os build of the raspberry pi 5 as its fairly new and also updating some drivers for the usb mic

After this, it was rather straightforward:

Picovoice was used for the voice commands, OpenAI libraries were used for the actual image generation and Tkinter was used for the User Frontend.